SPHEREx Image Quality and Alignment Verification

Technical Report | Task S1-13 follow-up: image quality, WCS overlap, and frame alignment

Key Results

Key metric	Result
Mutual star matches	18,873
Mean mutual-match error	0.2376 pixels
95th-percentile error	1.0 pixel
Maximum mutual-match error	2.0 pixels
Valid overlap pixels	3,924,348 (~94.3%)

1. Objective

This analysis evaluates the image quality of one SPHEREx Level 2 observation and performs a preliminary alignment verification using a second observation in the same spectral band (D3). The workflow examines FITS structure, image statistics, WCS footprints, reprojection, and star-position matching.

2. Data Used

First observation: 2025W19_1B_0116_1 (SPHEREx-D3).

Second observation: 2025W19_1B_0116_2 (SPHEREx-D3).

Both files are Level 2 SPHEREx DR1 products. Each FITS file contains seven HDUs: PRIMARY, IMAGE, FLAGS, VARIANCE, ZODI, EPSF, and WCS-WAVE.

The IMAGE arrays have dimensions 2040 × 2040 and data type float32. The second observation was selected from the IRSA SPHEREx archive and downloaded as a separate observation.

3. FITS Structure and Image Quality

The first observation's IMAGE, FLAGS, and VARIANCE extensions were inspected. IMAGE, FLAGS, and VARIANCE each have a 2040 × 2040 shape.

For the first IMAGE array, 4,161,367 of 4,161,600 pixels were finite (approximately 99.994%), while 233 pixels were NaN.

The first IMAGE statistics were: minimum −9997.67, maximum 48396.997, mean 0.3421, median 0.1802, and standard deviation 30.514. The large difference between the mean and median, together with the long right tail in the histogram, indicates a strongly right-skewed intensity distribution.

The 1st, 50th, 99th, and 99.9th percentiles were approximately 0.1176, 0.1802, 1.0772, and 15.5585, respectively. There were 3,316 negative finite pixels, including one pixel below −1000. There were 4,162 pixels above the 99.9th-percentile threshold.

Spatial plots showed that negative and high-value pixels were sparse rather than forming an obvious large contiguous region. These values were not removed. Their physical meaning should be interpreted using the product documentation and FLAGS definitions before creating a science-quality mask.

4. Second Observation and WCS Footprint

The second observation was selected with the same spectral band (D3) to make the comparison more consistent. Its IMAGE array also had shape 2040 × 2040 and data type float32, with 235 NaN pixels. Its reported minimum and maximum were −645.6845 and 49246.395.

The image-center coordinates derived from the WCS were approximately RA = 134.41458°, Dec = −9.00414° for the first observation, and RA = 134.35703°, Dec = −9.19229° for the second. The angular separation between the image centers was approximately 11.79 arcminutes.

Projecting the first image's corners into the second image's pixel coordinate system showed that the frames overlap partially. The visual footprint plot indicated that most of the second image is covered by the projected first image, with a strip near the

upper edge outside the overlap.

5. WCS Reprojection

The first IMAGE array was reprojected onto the pixel grid of the second observation using the two images' celestial WCS transformations and bilinear interpolation (order = 1). Pixels without sufficient valid source coverage were marked invalid.

The resulting reprojected array retained the 2040 × 2040 shape. The overlap mask contained 3,924,348 valid pixels, approximately 94.3% of the 4,161,600 pixels in the second image.

6. Star Detection and Matching

Bright local maxima were detected in the reprojected first image and the second image using a local-maximum filter and a threshold based on the median and median absolute deviation (MAD). The initial nearest-neighbor matching used a 5-pixel distance limit.

The initial matching found 19,851 candidate matches within 5 pixels. A one-to-one selection step retained 19,818 unique matches. Their median distance was 0 pixels, mean distance was approximately 0.3908 pixels, and maximum distance was approximately 4.4721 pixels.

A stricter mutual-nearest-neighbor check was then applied: a pair was retained only when each detection was the other's nearest neighbor and the separation was no greater than 2 pixels. This produced 18,873 mutual matches, with median error 0 pixels, mean error approximately 0.2376 pixels, 95th-percentile error 1 pixel, and maximum error 2 pixels.

7. Residual Analysis

For the 19,818 unique matches, the mean X and Y residuals were approximately −0.0144 and −0.0090 pixels. The median residual in each axis was 0 pixels, and the standard deviations were approximately 0.6457 pixels (X) and 0.6492 pixels (Y).

The 4 × 4 spatial analysis found matched stars in all 16 regions, with roughly 957–1,568 matches per region. Regional mean radial errors ranged approximately from 0.2825 to 0.4873 pixels, and regional 95th-percentile radial errors ranged approximately from 1.4142 to 2.4141 pixels.

The residual-vector plot did not show an obvious single global direction. However, the vectors were very small relative to the full image scale, so the plot was amplified for visualization only. The amplification did not alter the measured residuals.

8. Conclusion and Limitations

The two D3 observations have substantial WCS footprint overlap. Reprojection onto a common pixel grid yielded approximately 3.92 million valid overlap pixels. Star matching, especially the stricter mutual-nearest-neighbor check, produced a large set of consistent matches with a mean separation of approximately 0.238 pixels and a 95th-percentile separation of 1 pixel.

These results support a positive preliminary assessment of relative frame alignment after WCS reprojection. They do not, by themselves, constitute an independent absolute astrometric validation. The star detector uses integer-pixel local maxima, the matching procedure is based on nearest neighbors, and no external astrometric reference catalogue was used. A zero median can therefore reflect pixel-quantized peak positions rather than perfect subpixel registration.

The first-image quality inspection also identified NaNs and extreme values. No pixels were removed, and no definitive interpretation of individual FLAGS bits was asserted. Before scientific use, the flag-bit definitions and product-specific quality guidance should be checked against the relevant SPHEREx DR1 documentation.

Recommended next steps: (1) validate the matched sources against an external astrometric catalogue or trusted source list; (2) use centroid or PSF-fitting positions for subpixel accuracy; (3) inspect residuals as a function of image position and source brightness; and (4) consult the official FLAGS definitions before constructing a quality mask.

Appendix: Key Results

First observation: 2025W19_1B_0116_1 | Second observation: 2025W19_1B_0116_2 | Band: D3

Image dimensions: 2040 × 2040 | Valid overlap pixels: 3,924,348

Unique star matches: 19,818 | Mutual star matches: 18,873

Mutual-match median error: 0.0 pixels | Mean error: 0.2376 pixels | 95th percentile: 1.0 pixel | Maximum: 2.0 pixels